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'Project 4
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' {$STAMP BS2}
' {$PBASIC 2.5}

CM        CON 2260
CMD       VAR Word
time      VAR Word
row       VAR Nib
column    VAR Nib
keypad1   VAR Word
keypad2   VAR Word
temp      VAR Nib
constant  VAR Word
counter   VAR Byte
C1        VAR Byte
C2        VAR Byte
C3        VAR Byte
C4        VAR Byte
C5        VAR Byte
password1 VAR Bit(7)
password2 VAR Bit(7)
InPass    VAR Bit(7)

password1(0) = "1"
password1(1) = "2"
password1(2) = "3"
password1(3) = "4"
password1(4) = "5"
password1(5) = "A"
password1(6) = "#"

password2(0) = "A"
password2(1) = "5"
password2(2) = "4"
password2(3) = "3"
password2(4) = "2"
password2(5) = "1"
password2(6) = "*"

SEROUT 14, 84, [208]
SEROUT 14, 84, [22, 12, 17]
C1 = 0
FOR counter = 0 TO 5
PULSOUT 13, 250
PAUSE 10
NEXT
GOSUB start

start:
      counter = 0
      C3 = 0
      C4 = 0
      SEROUT 14, 84, [12, 128, "Enter Your Passcode:", 148]
      DO
      GOSUB READkeypad1
      IF keypad1 THEN
      IF keypad1 = keypad2 THEN
      PAUSE 50
      ELSE
      GOSUB display
      ENDIF
      ENDIF
      keypad2 = keypad1
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      IF (counter = 7 AND C2 = 0) THEN
      GOTO pswCheck1
      ENDIF
      LOOP
      RETURN

pswCheck1:
      counter = 0
      FOR counter = 0 TO 6
      IF (password1(counter) = InPass(counter)) THEN
      PAUSE 50
      ELSE
      FOR C2 = 0 TO 2
      SEROUT 14, 84, [12, 128, "Wrong Passcode!", 148, "Please Try Again"]
      PAUSE 700
      SEROUT 14, 84, [12]
      PAUSE 700
      NEXT
      counter = 0
      C2 = 0
      GOTO start
      ENDIF
      NEXT
      SEROUT 14, 84, [12]
      SEROUT 14, 84, [128, "Passcode Correct!!"]
      PAUSE 1500
      SEROUT 14, 84, [12]
      C2 = 0
      C5 = 0
      GOTO main
      RETURN

pswCheck2:
      LOW 9
      counter = 0
      FOR counter = 0 TO 6
      IF (password2(counter) = InPass(counter)) THEN
      PAUSE 50
      ELSE
      FOR C2 = 0 TO 2
      SEROUT 14, 84, [12, 128, "Wrong Passcode!", 148, "Please Try Again "]
      C1 = C1 + 1
      PAUSE 700
      PAUSE 700
      NEXT
      C2 = 0
      C4 = 0
      C5 = 0
      GOTO main
      ENDIF
      NEXT
      SEROUT 14, 84, [12, 128, "System Shutting Down"]
      PAUSE 1500
      counter = 0
      C2 = 0
      C3 = 0
      C5 = 0
      GOTO start
      RETURN

main:
      C4 = 0
      DO
      counter = 0
      GOSUB PING
      IF (CMD < 101 AND C2 = 0) THEN
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      C1 = 0
      GOSUB openDoor
      ELSEIF (CMD > 100 AND C2 = 1) THEN
      C1 = 0
      FOR counter = 1 TO 60
      GOSUB PING
      IF (CMD < 101) THEN
      C2 = 0
      GOTO main
      ENDIF
      NEXT
      GOSUB closeDoor
      ELSE
      IF (C2 = 0) THEN
      C4 = C4 + 1
      IF (C4 = 30) THEN
      SEROUT 14, 84, [168, "Enter Password to", 188, "shut down the system"]
      ENDIF
      GOSUB ReadKey
      ELSEIF (C2 = 1) THEN
      C1 = C1 + 1
      IF (C1 = 120) THEN
      C1 = 0
      C2 = 0
      GOTO obstacle
      ENDIF
      ENDIF
      ENDIF
      LOOP
      RETURN

READkeypad1:
      keypad1 = 0
      OUTL = %00000000
      DIRL = %00000000
      FOR row = 0 TO 3
      DIRB = %1111
      OUTB = %0000
      OUTA = 1 << row
      DIRA = 1 << row
      temp = 0
      FOR column = 0 TO 3
      INPUT (column + 4)
      temp = temp | (INB & (1 << column))
      NEXT
      keypad1 = keypad1 << 4 | (Temp REV 4)
      NEXT
      RETURN

ReadKey:
      GOSUB READkeypad1
      IF keypad1 THEN
      IF keypad1 = keypad2 THEN
      PAUSE 50
      ELSE
      GOSUB display
      ENDIF
      ENDIF
      keypad2 = keypad1
      RETURN

openDoor:
      LOW 9
      SEROUT 14, 84, [12, 148, "Door Opening"]
      FOR counter = 0 TO 42
      HIGH 8
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      PULSOUT 13, 1150
      PAUSE 10
      LOW 8
      PAUSE 10
      NEXT
      HIGH 8
      SEROUT 14, 84, [12, 148, "Door Open!"]
      C2 = 1
      PAUSE 500
      RETURN

closeDoor:
      LOW 8
      SEROUT 14, 84, [12, 148, "Door Closing"]
      FOR counter = 0 TO 18
      GOSUB PING
      IF (CMD < 101) THEN
      GOSUB openDoor
      RETURN
      ENDIF
      HIGH 9
      PULSOUT 13, 260
      PAUSE 10
      LOW 9
      PAUSE 10
      NEXT
      HIGH 9
      SEROUT 14, 84, [12, 148, "Door Closed"]
      C2 = 0
      PAUSE 500
      RETURN

obstacle:
      LOW 8
      LOW 9
      DO
      GOSUB PING
      IF (CMD > 100) THEN
      C2 = 1
      LOW 11
      LOW 10
      GOTO main
      ENDIF
      C2 = C2 + 1
      IF (C2 = 20) THEN
      C2 = 0
      IF (C1 = 0) THEN
      HIGH 11
      LOW 10
      SEROUT 14, 84, [218, 230]
      C1 = 1
      ELSE
      LOW 11
      HIGH 10
      SEROUT 14, 84, [219, 220]
      C1 = 0
      ENDIF
      ENDIF
      LOOP
      RETURN

kill:
      IF (IN12 = 1) THEN
      GOSUB openDoor
      LOW 11
      LOW 10
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      SEROUT 14, 84, [12, 133, "EMERGENCY!", 152, "DOOR IS OPEN"]
      DO WHILE (IN12 = 1)
      PAUSE 50
      LOOP
      PAUSE 10
      DO WHILE (IN12 = 0)
      PAUSE 50
      LOOP
      PAUSE 200
      GOSUB closeDoor
      SEROUT 14, 84, [12]
      GOTO main
      ENDIF
      RETURN

PING:
      PULSOUT 15, 5
      PULSIN 15, 1, time
      CMD = CM ** time
      GOSUB kill
      SEROUT 14, 84, [128, "Distance: ", 138, DEC3 CMD, 141, "cm"]
      RETURN

counting:
      IF (C4 > 29) THEN
      SEROUT 14, 84, [143, " "]
      C5 = C5 + 1
      counter = C5
      IF (C5 = 7) THEN
      GOSUB pswCheck2
      ENDIF
      ELSE
      counter = counter + 1
      ENDIF
      RETURN

Display:
      IF (C4 > 29) THEN
      counter = C5
      ENDIF
      IF keypad1.BIT15 THEN
      SEROUT 14, 84, ["1", 218, 220]
      InPass(counter) = "1"
      GOSUB counting
      PAUSE 7
      ELSEIF keypad1.BIT14 THEN
      SEROUT 14, 84, ["2", 218, 221]
      InPass(counter) = "2"
      GOSUB counting
      PAUSE 7
      ELSEIF keypad1.BIT13 THEN
      SEROUT 14, 84, ["3", 218, 222]
      InPass(counter) = "3"
      GOSUB counting
      PAUSE 7
      ELSEIF keypad1.BIT12 THEN
      SEROUT 14, 84, ["A", 218, 223]
      InPass(counter) = "A"
      GOSUB counting
      PAUSE 7
      ELSEIF keypad1.BIT11 THEN
      SEROUT 14, 84, ["4", 218, 224]
      InPass(counter) = "4"
      GOSUB counting
      PAUSE 7
      ELSEIF keypad1.BIT10 THEN
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SEROUT 14, 84, ["5", 218, 225]
InPass(counter) = "5"
GOSUB counting
PAUSE 7
ELSEIF keypad1.BIT9 THEN
SEROUT 14, 84, ["6", 218, 226]
InPass(counter) = "6"
GOSUB counting
PAUSE 7
ELSEIF keypad1.BIT8 THEN
SEROUT 14, 84, ["B", 218, 227]
InPass(counter) = "B"
GOSUB counting
PAUSE 7
ELSEIF keypad1.BIT7 THEN
SEROUT 14, 84, ["7", 218, 228]
InPass(counter) = "7"
GOSUB counting
PAUSE 7
ELSEIF keypad1.BIT6 THEN
SEROUT 14, 84, ["8", 218, 229]
InPass(counter) = "8"
GOSUB counting
PAUSE 7
ELSEIF keypad1.BIT5 THEN
SEROUT 14, 84, ["9", 218, 230]
InPass(counter) = "9"
GOSUB counting
PAUSE 7
ELSEIF keypad1.BIT4 THEN
SEROUT 14, 84, ["C", 218, 231]
InPass(counter) = "C"
GOSUB counting
PAUSE 7
ELSEIF keypad1.BIT3 THEN
SEROUT 14, 84, ["*", 219, 220]
InPass(counter) = "*"
GOSUB counting
PAUSE 7
ELSEIF keypad1.BIT2 THEN
SEROUT 14, 84, ["0", 219, 221]
InPass(counter) = "0"
GOSUB counting
PAUSE 7
ELSEIF keypad1.BIT1 THEN
SEROUT 14, 84, ["#", 219, 222]
InPass(counter) = "#"
GOSUB counting
PAUSE 7
ELSEIF keypad1.BIT0 THEN
SEROUT 14, 84, ["D", 219, 223]
InPass(counter) = "D"
GOSUB counting
PAUSE 7
ENDIF
RETURN
```